

ABERDARE TECHNICAL COLLEGE

ICT LEVEL 5 – MODULE 3

PRACTICAL 1 – OBSERVATION CHECKLIST

UNIT: NETWORK DESIGN AND MANAGEMENT

UNIT CODE: 0612 451 07A

ELEMENT 1: DESIGN COMPUTER NETWORK

DURATION: 3 HOURS

TOTAL PRACTICAL MARKS: 50

CANDIDATE DETAILS

Particular	Details
Candidate Name	_____
Admission Number	_____
Assessor Name	_____
Date	_____
Venue	_____
Candidate Signature	_____
Assessor Signature	_____

INSTRUCTIONS TO THE ASSESSOR

- ✓ *Ensure that each candidate has access to a functional computer with **Cisco Packet Tracer** installed.*
- ✓ *Provide the candidate with the required materials and ensure that the computer is working before the assessment begins.*
- ✓ *Brief the candidate on the assessment requirements without demonstrating the procedures or providing solutions.*
- ✓ *Ensure that the candidate saves the Packet Tracer file using the format **YourName_PracticalExam.pkt**.*
- ✓ *Observe the candidate throughout the practical and award marks only for work actually demonstrated or correctly completed.*
- ✓ *Assess the candidate's performance according to the items listed in the observation checklist.*
- ✓ *Do not award marks for a task that has not been completed or demonstrated correctly.*

CUTTING LIST

No.	Material	Quantity	Provided By
1	Desktop computer/laptop with Cisco Packet Tracer installed	1	Assessor
2	Word processing software	1	Assessor
3	Candidate tool	1	Assessor

PRACTICAL OBSERVATION CHECKLIST – 50 MARKS

SECTION A: STAR TOPOLOGY

TASK A1 – BUILD THE TOPOLOGY

No.	Items to be Evaluated	Allocated Marks	Candidate Score	Comments
A1.1	Places one 2960 switch and three PCs in the Packet Tracer workspace	1		
A1.2	Arranges the devices to represent a star topology	1		
A1.3	Connects PC1 to FastEthernet0/1 using copper straight-through cable	1		
A1.4	Connects PC2 to FastEthernet0/2 using copper straight-through cable	1		
A1.5	Connects PC3 to FastEthernet0/3 using copper straight-through cable	1		
A1.6	Confirms that the three links are green/up	1		
A1 Subtotal		6		

TASK A2 – IP CONFIGURATION

No.	Items to be Evaluated	Allocated Marks	Candidate Score	Comments
A2.1	Configures PC1 with 192.168.1.10 and subnet mask 255.255.255.0	2		
A2.2	Configures PC2 with 192.168.1.11 and subnet mask 255.255.255.0	1		
A2.3	Configures PC3 with 192.168.1.12 and subnet mask 255.255.255.0	1		
A2 Subtotal		4		

TASK A3 – VERIFICATION

No.	Items to be Evaluated	Allocated Marks	Candidate Score	Comments
A3.1	Executes ping from PC1 to 192.168.1.11	1		
A3.2	Executes ping from PC1 to 192.168.1.12	1		
A3.3	Confirms successful replies with 0% packet loss	1		
A3.4	Captures and saves the required screenshot	1		
A3 Subtotal		4		

TASK A4 – DISCUSSION TASK

No.	Items to be Evaluated	Allocated Marks	Candidate Score	Comments
A4.1	Explains why the three PCs must be in the same subnet for direct communication without a router	2		
A4.2	Describes what happens at Layer 2 when the subnet condition is not met	1		
A4 Subtotal		3		

SECTION A TOTAL: 17 MARKS

SECTION B: CONNECTING TWO NETWORKS WITH A ROUTER

TASK B1 – BUILD THE TOPOLOGY

No.	Items to be Evaluated	Allocated Marks	Candidate Score	Comments
B1.1	Places two switches and four PCs correctly	1		
B1.2	Connects two PCs to Switch A to form LAN A	1		
B1.3	Connects two PCs to Switch B to form LAN B	1		
B1.4	Places the 2911 router between the two LANs	1		
B1.5	Connects GigabitEthernet0/0 to Switch A correctly	1		
B1.6	Connects GigabitEthernet0/1 to Switch B correctly	1		
B1.7	Confirms all links are green/up	1		
B1 Subtotal		7		

TASK B2 – IP ADDRESSING SCHEME

No.	Items to be Evaluated	Allocated Marks	Candidate Score	Comments
B2.1	Configures LAN A PC1 with the correct IP address, subnet mask and gateway	2		
B2.2	Configures LAN A PC2 with the correct IP address, subnet mask and gateway	1		
B2.3	Configures LAN B PC1 with the correct IP address, subnet mask and gateway	2		
B2.4	Configures LAN B PC2 with the correct IP address, subnet mask and gateway	1		
B2 Subtotal		6		

TASK B3 – ROUTER CONFIGURATION VIA CLI

No.	Items to be Evaluated	Allocated Marks	Candidate Score	Comments
B3.1	Accesses the router CLI and selects GigabitEthernet0/0	1		
B3.2	Assigns 192.168.1.1 with subnet mask 255.255.255.0 to GigabitEthernet0/0	2		
B3.3	Enables GigabitEthernet0/0 using no shutdown	1		
B3.4	Configures GigabitEthernet0/1 with 192.168.2.1/24	2		
B3.5	Enables GigabitEthernet0/1 using no shutdown	1		
B3 Subtotal		7		

TASK B4 – VERIFICATION

No.	Items to be Evaluated	Allocated Marks	Candidate Score	Comments
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B4.1	Executes show ip interface brief correctly	1		
B4.2	Confirms GigabitEthernet0/0 is up/up	1		
B4.3	Confirms GigabitEthernet0/1 is up/up	1		
B4.4	Successfully pings a PC in LAN B from a PC in LAN A	2		
B4.5	Captures the required verification screenshots	1		
B4 Subtotal		6		

TASK B5 – TROUBLESHOOTING

No.	Items to be Evaluated	Allocated Marks	Candidate Score	Comments
B5.1	Uses show ip interface brief to identify the faulty interface	2		
B5.2	Correctly identifies the interface affected by the shutdown condition	1		
B5.3	Accesses the affected interface configuration mode	1		
B5.4	Restores the interface using the appropriate command	1		
B5.5	Verifies that the interface returns to up/up status	1		
B5.6	Records a clear description of how the fault was identified and corrected	1		
B5 Subtotal		8		

SECTION B TOTAL: 33 MARKS

MARK SUMMARY

Assessment Area	Marks
Section A – Star Topology	17
Section B – Two Networks with Router	33
TOTAL	50

FINAL SCORE

Candidate Score: _____ / 50

ASSESSOR'S COMMENTS

COMPETENCE DECISION

Competent: _____ **Not Yet Competent:** < _____

Assessor Name: _____

Signature: _____

Date: _____